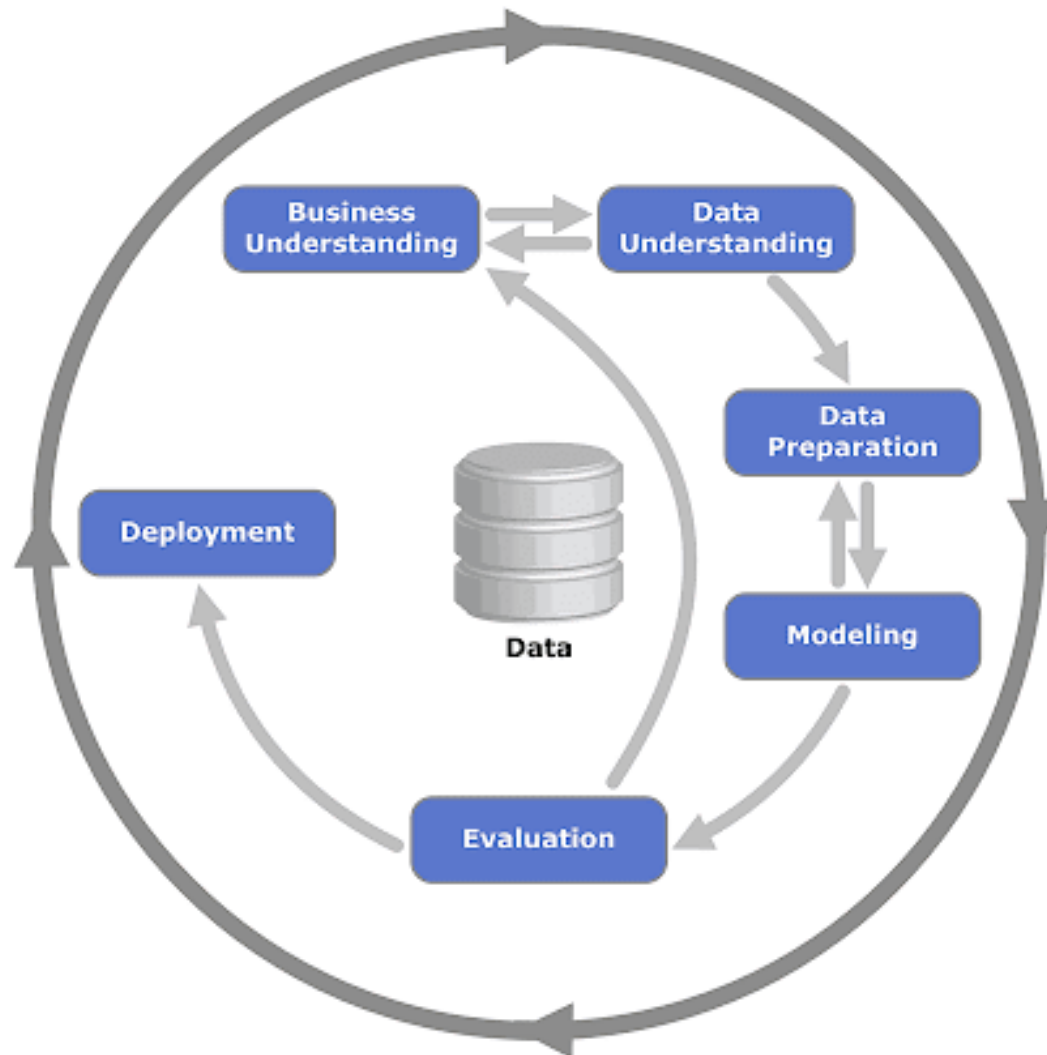




Data Visualization for Evaluation: Explaining to Stakeholders

CRISP-DM



Data Exploration

Purpose : For self or internal analysis

- Done up quickly. No need to be fanciful.
- To find patterns/trends/differences to get a sense of things
- To see which variables are correlated/clustered together
- Before Modelling

Data Visualization

Purpose : For management or external presentation

- Grab reader's attention
- Presented simply and clearly
- Compare data, highlights trends & differences
- After Modelling

Types of Visualizations

Comparison between variables

- Bar chart
- Box plot
- Bubble chart
- Histogram
- Line graph
- Stacked bar chart
- Radar chart
- Pie chart

Testing/viewing proportions

- Bubble chart
- Bubble map
- Stacked bar chart
- Word cloud

Relationship between variables

- Scatterplot
- Bar chart
- Radar chart
- Line graph
- Tree diagram

Types of Visualizations

Variable hierarchy

- Tree diagram
- Tree map

Data with locations

- Bubble map
- Geo mapping
- Dot map
- Flow map

Contribution analysis or part-to-whole

- When it is required to display constituents of a variable and contribution of each categorical level towards the overall variable, then the following charts are used:
 - Pie chart
 - Stacked bar chart
 - Donut chart

Types of Visualizations

Statistical distribution

- In order to understand the variation in a variable across different dimensions, represented by another categorical variable
- Box plot
- Bubble chart
- Histogram
- Stem and leaf plot

Unseen patterns

- For pattern recognition and relative importance of data points on different dimensions of a variable
- Bar chart
- Box plot
- Bubble chart
- Scatterplot
- Spiral plot
- Line chart

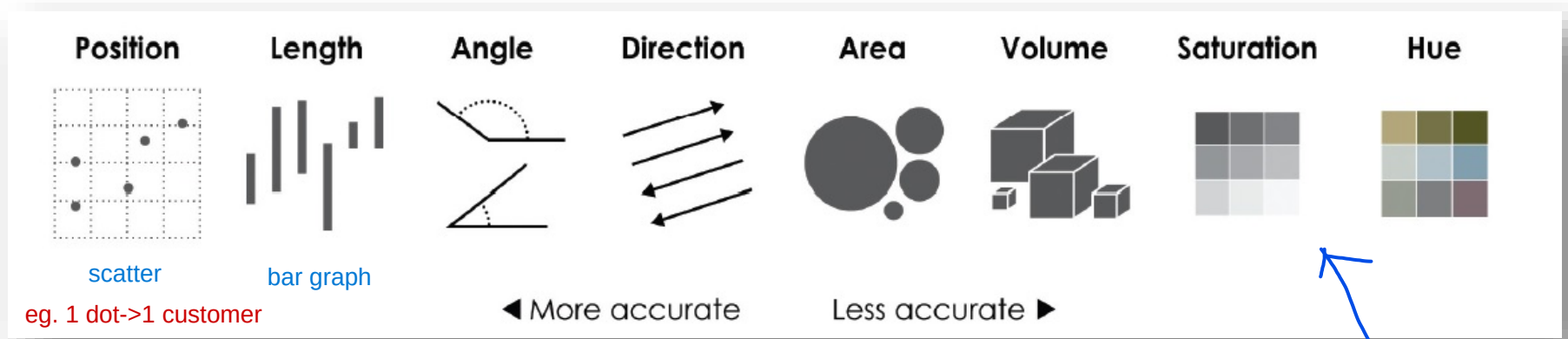
Spread of values or range:

- The following charts only give the spread of the data points across different bounds:
- Span chart
- Box plot
- Histogram

Textual data representation

- Word Cloud

Different visuals to represent the dimension/variable



Categorical	Ordinal	Quantitative
Position	Position	Position
□, ⊙, ○ Shape ✕, ✱	Size	Length
Different Colours	Colour Intensity	Size
	Different Colours	Colour Intensity
	Shape	

Key Principles

- **Time should always be on the horizontal axis** and the metric to be measured should always be on the vertical axis.
- Each graph should have a **name and label** so that the user does not have to go back to the data to understand what it is.
- Different visualizations for different types of audiences/stakeholders.

**WE  DATA
SCIENCE**